Elorie Bernard-Lacroix

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Education

Bachelor of Applied Science (BASc) in Engineering Science + PEY **Sept 2022 – Apr 2027**
University of Toronto *Toronto, ON*

- **Major/Minor:** Major in Machine Intelligence (AI/ML) Engineering, Minor in Environmental Engineering
- **Undergraduate Thesis:** "Multi-objective optimization of the reclamation of abandoned mine sites", applying machine learning and mathematical optimization to ecological remediation.
- **Non-Monetary Awards:** Dean's List (4 terms), Sustainability Scholar (2026 – Present)
- **Monetary Awards:** Procter & Gamble scholarship (2022 – 2026), Dean's Merit Award (2022), Faculty of Applied Science and Engineering Admission Scholarship (2022)

Experience

Software Engineering Intern **May 2025 – Aug 2026**
Procter & Gamble *North York, ON*

- Coordinated between operations teams, product teams, and consulting teams to develop and maintain **8 business-critical, production-grade** data validation pipelines between **3 applications** (Salesforce, Azure, SharePoint) using **Databricks** (Python, SQL), **Azure Data Factory**, and **Power BI**. Validated **2.5 million records daily** to consistently meet Service Level Agreements.
- Co-developed a full-stack web application with **FastAPI** (Python), **TypeScript**, **React**, and **SQLAlchemy**, containerized using **Docker**, and hosted on **GCP**. Designed unit tests and automated **CI/CD** with GitHub Actions.
- Developed and operated **7 web-scraping automations** in Python (Selenium, Pandas) with **multiprocessing** and **automated file tracking**, archiving **20 GB of data** and saving **$100,000** in vendor costs.
- Co-developed a proof-of-concept that uploads videos from **camera glasses** securely to a **private cloud** and performs **automated AI video analysis** (98% accuracy) to derive valuable **business insights**, while adhering to strict privacy policies.
- Built a pipeline that uses an **LLM** to classify up to **50,000 rows** of **unstructured data** per week (95% accuracy). Leveraged Arize for **AI observability** and continuous improvement.
- Co-developed an automation in Python that consolidates **10 dynamic sources of data** and provides automatic multi-page summaries of data alignment between two main systems, saving over **3,000 hours a year** and improving data accuracy for more than **70 users**.

Test Automation and Vehicle Instrumentation Intern **May 2024 – Aug 2024**
Bombardier Recreational Products (French-speaking Environment) *Valcourt, QC*

- Developed a **Python package** that automates a **battery test bench** in accordance with stakeholder needs and **IEC standards** (with technical documentation and flow charts), allowing for more precise testing and time savings.
- Interpreted existing technical documents and authored new **electrical schematics**, bills of materials, and **budgets** to troubleshoot and maintain **3 towing dynamometers** (trailers equipped with an Eddy current brake to apply a load to vehicles). Created a Google Sheets tool with **Apps Script** to facilitate the maintenance and operation of equipment.
- Led 2 main projects (above), **defining their scope** and **requirements**, with their respective tasks and deliverables.
- Led a training session with over **30 stakeholders**, demonstrating the operation of newly developed tools and ensuring **seamless integration** into existing workflows.

Information Technology Auditing Intern **May 2023 – Aug 2023**
Treasury Board Secretariat of the Ontario Public Service *Toronto, ON*

- Compiled **50 research extracts** on talent management in interactive Excel sheets with different views, including pivot tables and filtering options to assist the team in an upcoming audit.
- Used standards provided by **ISO, COBIT, NIST, and GO-ITS** to supplement research and guide audit recommendations.
- Recorded meeting notes for **client interviews**, managed documents, proofread the audit report, and performed research to assist in a 3-month audit of **End-User IT Asset Lifecycle Management**.

Information Technology Technician **Apr 2022 – Aug 2022**
Canada's Wonderland *Vaughan, ON*

- Collaborated with 3 other technicians to perform upgrades to **Simphony 2.0** and deploy over **300 POS systems** in the park, including their respective **peripherals** (scanners, printers, cameras) and required **peripheral drivers**, to ensure a fully operational cashless park.
- Troubleshot **networking**, **hardware**, and **software** issues for employees in the office.
- Configured **Cisco network phones** and **Cisco Finesse** (queues shared among phones) to allow for more effective online customer service support. Carefully **documented** this process for future technicians.

Projects

Canoe Generator (UofT Concrete Canoe Team) | *Python, Multi-Objective Optimization* **More Information**

- Worked with other Hull & Structural leads in the UofT Concrete Canoe design team to build a Python program that searches a given space of canoe parameters and optimizes for target values of drag, stability, and more.

Northern Ontario Graphite Mine Environmental Assessment | *Impact Assessment* **Final Report**

- Co-authored a partial Environmental Impact Assessment (EIA) for a proposed open-pit graphite mine in Ontario.
- Evaluated environmental, social, and economic Valued Components (VCs) using Leopold matrices to formulate mitigation and monitoring strategies for air quality, surface water, and local caribou habitats.

SmartStudy, AI Habit Optimizer | *Python, Machine Learning, GenAI, Bayesian Optimization* **More Information**

- Collaborated with two other students to build SmartStudy, a user-centric and data-driven habit optimizer that uses TabPFN and Bayesian optimization to predict and maximize student GPAs. Used k-Nearest Neighbors to surface peer-based behavioral insights and integrated GPT-4 to deliver natural language academic coaching. Also experimented with XGBoost, 1D CNN, and mixed models.

Solar-Powered Water Sanitation System | *Arduino, Fusion 360, 3D Printing* **More Information**

- Designed and prototyped an automated sanitation system that heats water to pasteurization temperature, maintains required dwell time, and dispenses it into a potable tank without manual intervention.
- Integrated a temperature sensor, a solar parabolic trough, and a motor-driven dispensing mechanism coordinated via custom Arduino firmware, with structural components fabricated via 3D printing.

Seam-Carving Program | *C, Dynamic Programming* **More Information**

- Designed a program that iteratively removes the least important parts of an image to resize it without visual distortion.

Predicting Failure Load | *MATLAB* **More Information**

- Built a MATLAB program that simulates a given bridge design and calculates its factor of safety against various forms of failure. Result: 3rd place.

Leadership & Community Involvement

Bikechain (non-profit, Do-It-Yourself bike shop) **Oct 2026 – Present**

Volunteer *Toronto, ON*

- Assisting cyclists with bike repairs, maintaining the lending bike fleet, and refurbishing/recycling donated bikes to promote sustainable transportation in the community.

Karma Grocery Store Co-op (non-profit, sustainable grocery store) **Sept 2026 – Present**

Active Member and Volunteer *Toronto, ON*

- Stocking shelves, labeling products, and assisting full-time employees with ad-hoc tasks to support daily operations and ultimately support the store's mission of offering ethically sourced food while reducing waste and supporting local growers.

University of Toronto Engineering Student Teams **Sept 2022 – Aug 2026**

Software Lead & Communications Manager *Toronto, ON*

- **Concrete Canoe Team:** Developed specialized Python software to search parameter spaces and optimize canoe designs for factors such as drag and stability.
- **Iron Dragons:** Served as Communications Manager coordinating logistics and fostering community engagement. Competed nationally with the team, securing a place at world championships.

Skills

Programming Languages: Python, C, Java, SQL, R, JavaScript/TypeScript, HTML/CSS
Data Analytics & AI: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, LLMs, Deep Learning, Power BI
Web Development & Frameworks: Flask, FastAPI, React, SQLAlchemy, Alembic (database migrations), PostgreSQL
Cloud Enablement & DevOps: Azure, Databricks, GCP, Git/GitHub, Docker, Linux, CI/CD (GitHub Actions), Unit Testing, Integration Testing, PyTest, Selenium
Infrastructure & Security: TCP/IP Networking, Database Management, Client Services Support, Access Management
Business & Process Enablement: Business Analysis, Process Automation, ServiceNow (ITSM), Stakeholder Training, IT Standards, Agile/Scrum, Jira, Confluence
Robotics & Hardware: Computer-Aided Design (Fusion 360, Onshape), Rapid Prototyping (3D Printing), Microcontrollers (Arduino, Raspberry Pi)